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Inventor(s)	Han; MiJin et al.

SLIDABLE DISPLAY DEVICE

Abstract

A slidable display device in one example includes a display panel assembly, a mounting frame, and a guide member tiltably connected to the mounting frame. The guide member can be configured to support the display panel assembly as the display panel assembly moves between a first position and a second position with respect to the mounting frame. The guide member can be configured to bend the display panel assembly as the display panel assembly moves from the first position to the second position so that a first portion of the display panel assembly is tilted downward.

Inventors: **Han; MiJin (Seoul, KR), Kim; Daeyun (Seoul, KR), Park; HoSung (Paju-si, KR)**

Applicant: **LG Display Co., Ltd.** (Seoul, KR)

Family ID: **1000008124334**

Assignee: **LG Display Co., Ltd. (Seoul, KR)**

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Korean Patent Application No. 10-2024-0027190 filed on Feb. 26, 2024, in the Korean Intellectual Property Office, the entire contents of which is hereby expressly incorporated by reference into the present application.

BACKGROUND

Field

[0002] The present disclosure relates to a display device, and more particularly, to a slidable display device which can display images even while sliding between a first position and a second position.

Discussion of the Related Art

[0003] Display devices employed as a monitor of a computer, a TV, a mobile phone or the like may include an organic light emitting display (OLED) that emits light by itself, and a liquid crystal display (LCD) that requires a separate light source.

[0004] Such display devices are being used in more and more applications, including computer monitors, televisions, and personal portable devices, as well as in vehicles. Accordingly, research is ongoing to develop display devices having a larger display area while reducing the volume of the display devices.

[0005] Further, a slidable display device is attracting attention as a next generation display device. Such a slidable display device is manufactured by forming a display part and lines on a flexible substrate made of a flexible material, such as plastic, so that it can display images even when it is expanded or contracted in a sliding manner. That is, the slidable display device can have a display area that is active while it is sliding between an exposed state and a retracted state.

SUMMARY OF THE DISCLOSURE

[0006] An object to be achieved by the present disclosure is to provide a slidable display device mounted on a roof panel of a vehicle to display images.

[0007] Another object to be achieved by the present disclosure is to provide a slidable display device whose display panel is extended out of the inside of a roof panel to display images.

[0008] Yet another object to be achieved by the present disclosure is to provide a slidable display device whose disposition angle can be controlled in a state in which a display panel is slid.

[0009] Objects of the embodiments of the present disclosure are not limited to the above-mentioned objects, and other objects, which are not mentioned above, can be clearly understood by those skilled in the art from the following descriptions.

[0010] According to an aspect of the present disclosure, a slidable display device comprises a mounting frame, a display panel assembly movably coupled to the mounting frame, and a guide member which is tiltably coupled to the mounting frame, supports the display panel assembly, and guides movement of the display panel assembly, wherein the guide member is configured to bend at least a part of the display panel assembly while being tilted.

[0011] Other detailed matters of the embodiments are included in the detailed description and the drawings.

[0012] According to the present disclosure, a slidable display device mounted on a roof panel of a vehicle can display images on a portion of the display device extending to the inside of the vehicle while it is slid.

[0013] According to the present disclosure, the slidable display device can display images while it is being extended from inside to outside the roof panel.

[0014] According to the present disclosure, the slidable display device can display images at a position desired by a user by adjusting a take-out angle of a display panel assembly.

[0015] According to the present disclosure, the stability of the display panel assembly can be secured while the slidable display device is slid.
[0016] According to the present disclosure, the durability of the display panel assembly can be enhanced while the slidable display device is not used.
[0017] The effects of the present disclosure are not limited to the aforementioned effects, and other effects, which are not mentioned above, will be apparently understood to a person having ordinary skill in the art from the following description.
[0018] The objects to be achieved by the present disclosure, the means for achieving the objects, and the effects of the present disclosure described above do not specify essential features of the claims, and, thus, the scope of the claims is not limited to the disclosure of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0019] The above and other aspects, features and other advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:
[0020] FIG. **1**A is a perspective view schematically illustrating a slidable display device mounted on a roof panel of a vehicle according to an embodiment of the present disclosure;
[0021] FIG. **1**B is a bottom perspective view schematically illustrating the slidable display device mounted on the roof panel of the vehicle according to an embodiment of the present disclosure;
[0022] FIG. **2**A is a perspective view illustrating a structure of the slidable display device according to an embodiment of the present disclosure;
[0023] FIG. **2**B is a bottom perspective view illustrating the structure of the slidable display device according to an embodiment of the present disclosure;
[0024] FIG. **3** is a side view of the slidable display device according to an embodiment of the present disclosure;
[0025] FIG. **4**A is a side view illustrating a structure of a guide member according to an embodiment of the present disclosure;
[0026] FIG. **4**B is a side view illustrating a guide driver being tilted according to an embodiment of the present disclosure;
[0027] FIG. **4**C is a side view of the guide support being deployed according to an embodiment of the present disclosure;
[0028] FIG. **5** is an exploded perspective view of the slidable display device according to an embodiment of the present disclosure;
[0029] FIG. **6**A is a perspective view of a guide member according to an embodiment of the present disclosure;
[0030] FIG. **6**B is a perspective view illustrating a guide driver which is bent according to an embodiment of the present disclosure;
[0031] FIG. **6**C is an enlarged perspective view illustrating a portion where a guide body is connected to an opening and closing plate according to an embodiment of the present disclosure;
[0032] FIG. **7**A is a bottom perspective view of a guide support which is deployed according to an embodiment of the present disclosure;
[0033] FIG. **7**B is an exploded perspective view of the guide support according to an embodiment of the present disclosure;
[0034] FIG. **8**A is a perspective view illustrating a structure of a panel transfer unit according to an embodiment of the present disclosure;
[0035] FIG. **8**B is an exploded perspective view illustrating the structure of the panel transfer unit according to an embodiment of the present disclosure;

[0036] FIG. **9** is an exploded perspective view of a display panel assembly according to an embodiment of the present disclosure;
[0037] FIG. **10**A is a perspective view of the display panel assembly which is accommodated according to an embodiment of the present disclosure;
[0038] FIG. **10**B is a bottom perspective view of the display panel assembly which is accommodated according to an embodiment of the present disclosure;
[0039] FIG. **11**A is a perspective view of the display panel assembly which is bent according to an embodiment of the present disclosure;
[0040] FIG. **11**B is a bottom perspective view of the display panel assembly which is bent according to an embodiment of the present disclosure;
[0041] FIG. **12**A is a perspective view illustrating that a display panel of the display panel assembly is moved and exposed to the outside of a roof panel according to an embodiment of the present disclosure; and
[0042] FIG. **12**B is a bottom perspective view illustrating that the display panel of the display panel assembly is moved and exposed to the outside of the roof panel according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0043] The embodiments described herein can be variously modified. Specific embodiments are described in the drawings and can be described in detail in the detailed description. It should be understood, however, that the specific embodiments disclosed in the accompanying drawings are only intended to facilitate an understanding of the various embodiments of the present disclosure. Therefore, it is to be understood that the technical idea of the present disclosure is not limited by the specific embodiments disclosed in the accompanying drawings, but the present disclosure should be understood to include all equivalents or alternatives included in the spirit and technical scope of the present disclosure.
[0044] Terms including ordinals, such as first, second, etc., can be used to describe various elements, but such elements are not limited to the above terms. The above terms are used only for the purpose of distinguishing one component from another.
[0045] It will be further understood that the terms “comprises” and/or “comprising,” or “includes” and/or “including,” or “has” and/or “having” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, components and/or groups thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components and/or groups thereof. When a component is “linked”, “coupled”, or “connected” to another component, the component can be directly linked or connected to the other component. However, unless specifically stated otherwise, it should be understood that a third component can be interposed between the components which can be indirectly linked or connected.
[0046] A “module” or a “unit” for control elements used in the present specification performs at least one function or operation. Further, the “module” or the “unit” can perform functions or operations by hardware, software, or a combination of hardware and software. Further, a plurality of “modules” or a plurality of “units”, other than a “module” or a “unit” that is to be performed in specific hardware or performed in at least one processor, can be integrated into at least one module. The singular forms “a,” “an,” and “the” include plural expressions unless the context clearly dictates otherwise.
[0047] The use of “exemplary” in the present disclosure means to serve as an example as opposed to identifying a preferred embodiment.
[0048] Besides, in the following description of the present disclosure, the detailed description of known functions and configurations incorporated herein will be omitted when it can make the subject matter of the present disclosure rather unclear.
[0049] Hereinafter, various embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. All the components of each slidable display device

according to all embodiments of the present disclosure are operatively coupled and configured.

[0050] FIG. **1**A is a schematic perspective view illustrating a slidable display device mounted on a roof panel of a vehicle according to an embodiment of the present disclosure. FIG. **1**B is a bottom perspective view schematically illustrating the slidable display device mounted on the roof panel of the vehicle according to an embodiment of the present disclosure. FIG. **2**A is a perspective view illustrating a structure of the slidable display device according to an embodiment of the present disclosure. FIG. **2**B is a bottom perspective view illustrating the structure of the slidable display device according to an embodiment of the present disclosure. FIG. **3** is a side view of the slidable display device according to an embodiment of the present disclosure. FIG. **4**A is a side view illustrating a structure of a guide member according to an embodiment of the present disclosure. FIG. **4**B is a side view illustrating a guide driver being tilted according to an embodiment of the present disclosure. FIG. **4**C is a side view of the guide support being deployed according to an embodiment of the present disclosure. FIG. **5** is an exploded perspective view of the slidable display device according to an embodiment of the present disclosure.

[0051] Referring to FIG. **1**A through FIG. **5**, a structure of a slidable display device **1000** according to an embodiment of the present disclosure will be described.

[0052] The slidable display device **1000** according to an embodiment of the present disclosure can be installed between an outer plate of a vehicle and a roof panel **10** of the vehicle disposed adjacent to the outer plate and under the outer plate.

[0053] The roof panel **10** of the vehicle includes an opening **11** penetrating in a vertical direction (see FIG. **1**A). The opening **11** has a predetermined width along one direction along which the slidable display device **1000** extends (see FIG. **1**A) and extends along an intersection direction (orthogonal to the one direction in a horizontal plane) (see FIG. **1**A).

[0054] A display panel assembly **1500** of the slidable display device **1000** can be accommodated between the outer plate and the roof panel **10** of the vehicle or can slide and extend to the inside of the vehicle through the opening **11**.

[0055] Further, the roof panel **10** of the vehicle includes slits **12** respectively extending from both sides of the opening **11** along the one direction.

[0056] The slidable display device **1000** can also be applied to other objects with roofs in addition to the roof panel **10** of the vehicle.

[0057] The slidable display device **1000** includes a mounting frame **1100**, a guide member **1200**, a roller unit **1300**, a panel transfer unit **1400**, and the display panel assembly **1500**.

[0058] The mounting frame **1100** is disposed between the outer plate of the vehicle and the roof panel **10**. Further, the guide member **1200**, the roller unit **1300**, and the panel transfer unit **1400** are mounted on the mounting frame **1100**.

[0059] Referring to FIG. **1**A through FIG. **2**B, the mounting frame **1100** includes at least one first support frame **1110**, a second support frame **1120**, and at least one fixing bracket **1130**.

[0060] A plurality (e.g., a pair) of first support frame **1110** is provided and spaced apart from each other on the roof panel **10** along the intersection direction. Each of the plurality of first support frames **1110** disposed on the roof panel **10** is supported on the roof panel **10** by the fixing bracket **1130**.

[0061] Further, an end of the first support frame **1110** is coupled to a guide body **1210** of the guide member **1200**.

[0062] Further, one end of the first support frame **1110** has a rounded surface at an upper portion thereof.

[0063] The second support frame **1120** is mounted on the roof panel **10** and serves to connect the plurality of first support frames **1110**. The first support frame **1110** is supported on the second support frame **1120**. The second support frame **1120** extends in the intersection direction and has a plate shape having a predetermined width along the one direction.

[0064] The fixing bracket **1130** is installed on the roof panel **10** and supports the plurality of first support frames **1110**. A plurality of fixing brackets **1130** is provided and fixed to outer surfaces of the plurality of first support frames **1110**. A part of the fixing bracket **1130** extends in the vertical direction (see FIG. **1**A) and the other part extends in the intersection direction.

[0065] A mounting bracket configured to fix the first support frame **1110** to the outer plate of the vehicle can be included in the mounting frame **1100**. The mounting bracket can be fixed to an inner surface of the first support frame **1110**. A part of the mounting bracket can extend in the vertical direction and another part can extend in the intersection direction.

[0066] FIG. **6**A is a perspective view of a guide member according to an embodiment of the present disclosure. FIG. **6**B is a perspective view illustrating a guide driver which is bent according to an embodiment of the present disclosure. FIG. **6**C is an enlarged perspective view illustrating a portion where a guide body is connected to an opening and closing plate according to an embodiment of the present disclosure.

[0067] Referring to FIG. **1**A through FIG. **6**C, the guide member **1200** is tiltably coupled to the mounting frame **1100**, supports the display panel assembly **1500**, and guides movement of the display panel assembly **1500**. The guide member **1200** is configured to bend at least a part of the display panel assembly **1500** while the guide member **1200** is being tilted.

[0068] The guide member **1200** is configured to be tilted between an angle at which the display panel assembly **1500** is horizontally disposed and an angle at which a part of the display panel assembly **1500** is extended below the mounting frame **1100**.

[0069] The guide member **1200** includes at least one guide body **1210**, a guide driver **1220**, a guide support **1230**, an opening and closing plate **1240**, and a sensor **1250**.

[0070] A plurality of guide bodies **1210** is provided, disposed at an end of the first support frame **1110**, and coupled to an outer side surface of the first support frame **1110**. A lower portion of the guide body **1210** is disposed in the slit **12** of the roof panel **10**. At least one of the guide bodies **1210** is equipped with the guide driver **1220** and the sensor **1250**.

[0071] Further, the guide body **1210** includes support protrusions **1211** and insertion protrusions **1212** to rotatably support the opening and closing plate **1240**. The support protrusions **1211** and the insertion protrusions **1212** of the plurality of guide bodies **1210** rotatably support opposite sides of the opening and closing plate **1240**.

[0072] A part of the support protrusion **1211** extends from a lower portion of the guide body **1210** along the intersection direction, and the other part extends along the one direction and protrudes toward the front of the first support frame **1110**. The support protrusions **1211** are disposed in the opening **11**. For example, ends of the support protrusions **1211** can extend in the opening **11**.

[0073] The insertion protrusion **1212** protrudes along the intersection direction in the other part of the support protrusion **1211**. The insertion protrusions **1212** are inserted into grooves formed in the opening and closing plate **1240**. The opening and closing plate **1240** is rotatably coupled to the support protrusions **1211** by the insertion protrusions **1212**.

[0074] The support protrusions **1211** disposed in the opening **11** support the opening and closing plate **1240** through the insertion protrusions **1212**. Therefore, when the opening and closing plate **1240** is not rotated by the guide driver **1220**, the opening and closing plate **1240** is maintained horizontally with respect to the roof panel **10**. Thus, the aesthetics of the interior of the vehicle can be improved. Further, the support protrusions **1211** support the opening and closing plate **1240** at positions where the support protrusions **1211** protrude toward the front of the first support frame **1110**. Thus, the opening and closing plate **1240** can be rotated without interference with the first support frame **1110**.

[0075] The guide driver **1220** is installed on the mounting frame **1100**, operatively connected to the guide body **1210** and rotatably supports the guide support **1230**.

[0076] The guide driver **1220** includes a motor **1221**, at least one link assembly **1222**, and a guide belt **1223**.

[0077] The motor **1221** is installed on the other side of the mounting frame **1100** and generates a

rotational driving force. The motor **1221** is operatively connected to the guide belt **1223** and transmits the rotational driving force to the guide belt **1223**.
[0078] The motor **1221** includes a motor pulley **1221***a* and a motor support member **1221***b*.
[0079] The guide belt **1223** is wound around the motor pulley **1221***a*.
[0080] The motor support member **1221***b* is installed on the second support frame **1120** and rotatably supports the motor **1221** and the motor pulley **1221***a*. One side surface of the motor support member **1221***b* is inwardly recessed, and the motor support member **1221***b* guides a part of the guide belt **1223** wound around the motor pulley **1221***a*. For example, a part of the guide belt **1223** can move within the one side surface of the motor support member **1221***b*.
[0081] The motor **1221** can be connected to a power generator, such as an external power supply or a built-in battery and supplied with power. The motor **1221** generates a rotational driving force and transmits the rotational driving force to the motor pulley **1221***a*.
[0082] When the motor **1221** is supplied with power from the power generator, the motor **1221** generates a rotational driving force. Further, when a plurality of sensors **1250** senses a rotation link **1222***ba*, the generation of the rotational driving force by the motor **1221** can be stopped in response to signals transmitted from at least one sensor **1250** to the power generator. This will be described in more detail along with the sensor **1250**.
[0083] The link assembly **1222** is rotatably installed on the guide body **1210** and supports the guide support **1230** and the opening and closing plate **1240**. A plurality of link assemblies **1222** is provided and installed on the plurality of guide bodies **1210**, respectively. The link assemblies **1222** are provided on opposites sides of the mounting frame **1100**.
[0084] One of the link assemblies **1222** (a link assembly located on the left of the drawing) receives the rotational driving force generated by the motor **1221** through the guide belt **1223**. Then, the one link assembly **1222** tilts (rotates) the guide support **1230** and the opening and closing plate **1240**. The other link assembly **1222** can assist in stably supporting the guide support **1230** and the opening and closing plate **1240** as they tilt (rotate).
[0085] The link assembly **1222** includes a link pulley **1222***a*, a first link **1222***b*, a second link **1222***c*, and a third link **1222***d*.
[0086] The link pulley **1222***a* is rotatably coupled to the guide body **1210** and the first support frame **1110**, and the guide belt **1223** is wound around the link pulley **1222***a*. The link pulley **1222***a* is coaxially coupled to one side of the rotation link **1222***ba* (and the other side of a connection link **1222***bb*).
[0087] The first link **1222***b* tilts the guide support **1230** by means of the rotational driving force transmitted from the guide belt **1223**.
[0088] The first link **1222***b* includes the rotation link **1222***ba*, the connection link **1222***bb*, and a support link **1222***bc*.
[0089] One side of the rotation link **1222***ba* is rotatably installed on the guide body **1210** and coaxially coupled to the link pulley **1222***a*. The rotation link **1222***ba* is tilted between a first position and a second position, and its rotation range is restricted by the plurality of sensors **1250**.
[0090] The first position refers to a disposition angle at which the display panel assembly **1500** is horizontally disposed in the roof panel **10**. Further, the second position refers to a disposition angle at which a part of the display panel assembly **1500** is extended below the mounting frame **1100**. Specifically, the first position and the second position are disposed along a circumferential direction of a circle centering on the one side of the rotation link **1222***ba* (or the intersection direction) as a center axis. The second position is disposed higher than the first position.
[0091] A tilting angle of the rotation link **1222***ba* can be restricted by one of a first sensor **1251** and a second sensor **1252**, which will be described later.
[0092] One side of the connection link **1222***bb* is coupled to one side of the rotation link **1222***ba* and the link pulley **1222***a* and the other side thereof is fixed to the support link **1222***bc*. The connection link **1222***bb* transmits, to the support link **1222***bc*, the rotational driving force the guide

belt **1223** which has been transmitted through the rotation link **1222***ba*.
[0093] The connection link **1222***bb* is tilted with respect to the rotation link **1222***ba*. The connection link **1222***bb* is disposed at an obtuse angle with respect to the rotation link **1222***ba*.
[0094] Further, one side of the third link **1222***d* is rotatably coupled to a central portion of the connection link **1222***bb*. While the connection link **1222***bb* is tilted, the third link **1222***d* is applied with pressure by the connection link **1222***bb*. For example, tilting of the connection link **1222***bb* imparts a rotational force to an end of the third link **1222***d*.
[0095] The support link **1222***bc* supports the guide support **1230**. One side of the support link **1222***bc* is fixed to the connection link **1222***bb* and the other side thereof is fixed to a movable block **1231***c* of the guide support **1230**. While the connection link **1222***bb* is tilted, the support link **1222***bc* tilts the guide support **1230**. Further, while the guide support **1230** is tilted, the display panel assembly **1500** supported on the guide support **1230** is bent. The support link **1222***bc* has a plate shape extending in the intersection direction.
[0096] The second link **1222***c* is installed on the opening and closing plate **1240** and applies pressure to the opening and closing plate **1240** in conjunction with the first link **1222***b* and the third link **1222***d*. While the second link **1222***c* is disposed vertically with respect to (e.g., normal to) the opening and closing plate **1240**, one side of the second link **1222***c* is coupled to the opening and closing plate **1240**. The other side of the second link **1222***c* is rotatably coupled to the third link **1222***d*.
[0097] The third link **1222***d* connects the first link **1222***b* and the second link **1222***c*. With the one side of the connection link **1222***bb* and the support protrusions **1211** as a rotation stage, the first link **1222***b*, the second link **1222***c*, and the third link **1222***d* constitute a pivotable four-bar link structure. Further, the first link **1222***b*, the second link **1222***c*, and the third link **1222***d* are rotated relative to each other to tilt (rotate) the guide support **1230** and the opening and closing plate **1240**.
[0098] The guide belt **1223** is disposed inside the first support frame **1110** in the intersection direction and transmits the rotational driving force generated by the motor **1221** to the link assembly **1222**. One side of the guide belt **1223** is wound around the motor pulley **1221***a* and the other side thereof is wound around the link pulley **1222***a* of the link assembly **1222**. Thus, the guide belt **1223** can be located between the first support frame **1110** and the motor support **1221***b*.
[0099] The guide support **1230** is connected to the display panel assembly **1500** and tilted with respect to the mounting frame **1100**. An end of the guide support **1230** is coupled to an end of the display panel assembly **1500**. The guide support **1230** is applied with pressure by the display panel assembly **1500** which is moved along the one direction by the panel transfer unit **1400**. As the guide support **1230** increases in length, as described below, the guide support **1230** guides sliding of the display panel assembly **1500**.
[0100] The guide support **1230** has a telescopic structure which can be increased in length. The increased length of the guide support **1230** can be three times to four times greater than the length of the first support frame **1110** along the vertical direction (i.e., the height of the mounting frame **1100**). While the increased length of the guide support **1230** is described as being up to three times to four times greater than the length of the first support frame **1110** along the vertical direction, it is not so limited. However, an excessive stress could be generated in the guide support **1230** by a load of the extended guide support **1230** if it is increased too much, which could result in a decrease in durability of the guide support **1230**.
[0101] FIG. **7**A is a bottom perspective view of a guide support which is deployed according to an embodiment of the present disclosure. FIG. **7**B is an exploded perspective view of the guide support according to an embodiment of the present disclosure.
[0102] Referring to FIG. **1**A through FIG. **7**B, the guide support **1230** includes a main guide plate **1231**, a plurality of sub-guide plates **1232** and **1233**, and a connection guide plate **1234**.
[0103] The main guide plate **1231** is coupled to the support link **1222***bc*. Both sides of the main guide plate **1231** are supported by a plurality of support links **1222***bc* of the guide driver **1220**.

[0104] An accommodation groove **1231**_a_ is inwardly recessed at an end on one side of the main guide plate **1231**. In the accommodation groove **1231**_a_ of the main guide plate **1231**, the plurality of sub-guide plates **1232** and **1233** is accommodated to be deployed in a parallel direction with respect to the main guide plate **1231**.

[0105] The main guide plate **1231** includes at least one movable rail **1231**_b_, the movable block **1231**_c_, at least one guide roller **1231**_d_, at least one guide support roller **1231**_e_, at least one auxiliary movable rail **1231**_f_, and at least one auxiliary movable block **1231**_g_.

[0106] A plurality of movable rails **1231**_b_ is provided and installed at both upper ends, respectively, of the main guide plate **1231**. The movable rail **1231**_b_ extends parallel with respect to the main guide plate **1231**. The movable rail **1231**_b_ serves as a path for sliding the movable block **1231**_c_.

[0107] A plurality of movable blocks **1231**_c_ is provided and relatively movably coupled to the plurality of movable rails **1231**_b_, respectively, in a sliding manner.

[0108] While the movable rail **1231**_b_ is fixed, the movable block **1231**_c_ can move forwards or backwards on the movable rail **1231**_b_. Meanwhile, while the movable block **1231**_c_ is fixed, the movable rail **1231**_b_ can move forwards or backwards.

[0109] Further, the support link **1222**_bc_ is fixed onto the movable block **1231**_c_.

[0110] Before the guide support **1230** is tilted (i.e., while the guide support **1230** is horizontally disposed), the movable block **1231**_c_ is disposed on the other side of the movable rail **1231**_b_ as shown in FIG. **4**A. Further, while the guide support **1230** is tilted, the movable block **1231**_c_ is disposed adjacent to one side of the movable rail **1231**_b_ as shown in FIG. **4**B.

[0111] When the guide driver **1220** operates to tilt the guide support **1230**, the movable rail **1231**_b_ relatively movably coupled to the movable block **1231**_c_ is pushed in conjunction with tilting (rotation) of the link assembly **1222**. Since the movable rail **1231**_b_ relatively movably coupled to the movable block **1231**_c_ is pushed in conjunction with tilting (rotation) of the link assembly **1222**, the movable rail **1231**_b_ is tilted together with the guide support **1230** while maintaining the same radius based on a bending portion of the display panel assembly **1500**.

[0112] A plurality of guide rollers **1231**_d_ is provided and rotatably installed at both ends on the other side of the main guide plate **1231**. The guide roller **1231**_d_ is rotatably mounted on the first support frame **1110**. For example, the guide roller **1231**_d_ can be mounted on the rounded surface of the first support frame **1110**. When the guide support **1230** is tilted, the guide roller **1231**_d_ moves along the rounded surface of the first support frame **1110** and supports the main guide plate **1231** to stably tilt the main guide plate **1231**.

[0113] A plurality of guide support rollers **1231**_e_ is provided and protrudes from a central portion on the other side of the main guide plate **1231** to be located below the main guide plate **1231**. Further, the plurality of guide support rollers **1231**_e_ comes into rotational contact with an upper portion of the display panel assembly **1500**. When the guide support **1230** is tilted, the plurality of guide support rollers **1231**_e_ is disposed to face a second lower roller **1322**, which will be described later. While the guide support **1230** is tilted, each of upper and lower sides of the bending portion of the display panel assembly **1500** is supported by the plurality of guide support rollers **1231**_e_ and a plurality of second lower rollers **1322**. Thus, the display panel assembly **1500** can stably slide. Therefore, it is possible to suppress damage to the display panel assembly **1500** while the display panel assembly **1500** slides.

[0114] A plurality of auxiliary movable rails **1231**_f_ is provided and installed on both inner surfaces of the main guide plate **1231**, which form the accommodation groove **1231**_a_. The auxiliary movable rail **1231**_f_ extends parallel with respect to the main guide plate **1231**. The auxiliary movable rail **1231**_f_ serves as a path for sliding the auxiliary movable block **1231**_g_.

[0115] A plurality of auxiliary movable blocks **1231**_g_ is provided and slidably coupled to the plurality of auxiliary movable rails **1231**_f_, respectively. The auxiliary movable block **1231**_g_ can be pushed or pulled parallel with respect to the main guide plate **1231** and moved on the auxiliary movable rail **1231**_f_ in conjunction with sliding of the display panel assembly **1500**. A sub-guide plate (i.e., a first sub-guide plate **1232**) is coupled to the auxiliary movable block **1231***g*. While the auxiliary movable block **1231***g* moves forwards and backwards along the auxiliary movable rail **1231***f*, the sub-guide plate (i.e., the first sub-guide plate **1232**) can be moved forwards (i.e., deployed) or moved backwards (i.e., accommodated).

[0116] A plurality of sub-guide plates **1232** and **1233** is installed in the accommodation groove **1231***a* of the main guide plate **1231**. The plurality of sub-guide plates **1232** and **1233** can be deployed parallel with respect to the main guide plate **1231**. The plurality of sub-guide plates **1232** and **1233** includes driving rails **1232***a* and **1233***a* and driving blocks **1232***b* and **1233***b*, respectively, and overlaps each other to be deployed in a telescopic structure.

[0117] In an embodiment of the present disclosure, the number of the plurality of sub-guide plates **1232** and **1233** is two as shown in FIG. **7**A and FIG. **7**B. Further, the two sub-guide plates **1232** and **1233** include the first sub-guide plate **1232** and a second sub-guide plate **1233**. Further, the driving rails for the two sub-guide plates **1232** and **1233** include a first driving rail **1232***a* and a second driving rail **1233***a*. Furthermore, the driving blocks for the two sub-guide plates **1232** and **1233** include a first driving block **1232***b* and a second driving block **1233***b*.

[0118] The first sub-guide plate **1232** is a sub-guide plate having a greater size among the plurality of sub-guide plates **1232** and **1233**. The inside of the first sub-guide plate **1232** is hollowed (e.g., recessed) along a direction along which the main guide plate **1231** extends. The second sub-guide plate **1233** is accommodated in the hollow portion of the first sub-guide plate **1232**.

[0119] The first driving rail **1232***a* and the first driving block **1232***b* are installed in the hollow inside of the first sub-guide plate **1232**.

[0120] The first driving rail **1232***a* is installed on an inner surface of the first sub-guide plate **1232**. The first driving rail **1232***a* extends parallel with respect to the first sub-guide plate **1232**. The first driving block **1232***b* is slidably coupled to the first driving rail **1232***a*.

[0121] The first driving block **1232***b* is slidably coupled to the first driving rail **1232***a*. The first driving block **1232***b* can be pushed or pulled parallel with respect to the first sub-guide plate **1232** and moved on the first driving rail **1232***a* in conjunction with sliding of the display panel assembly **1500**.

[0122] The second sub-guide plate **1233** is fixed to the first driving block **1232***b*. Further, the second sub-guide plate **1233** can be accommodated in the first sub-guide plate **1232** or deployed to the outside of the first sub-guide plate **1232** in conjunction with forward and backward movements of the first driving block **1232***b*.

[0123] The first driving rail **1232***a* and the first driving block **1232***b* can be configured as linear motion guides.

[0124] Further, a first guide block **1232***c* is installed on an outer side surface of the first sub-guide plate **1232**.

[0125] The first guide block **1232***c* engages with the movable rail **1231***b* at a position where the first guide block **1232***c* does not interfere with the movable block **1231***c*. The first guide block **1232***c* includes a slit inwardly recessed in one surface facing the movable rail **1231***b*. The first guide block **1232***c* is slidably coupled to the movable rail **1231***b* through the slit. The first sub-guide plate **1232** can be stably deployed from the main guide plate **1231** or accommodated into the main guide plate **1231** by the first guide block **1232***c*.

[0126] The second sub-guide plate **1233** is a sub-guide plate having a smaller size among the plurality of sub-guide plates **1232** and **1233**. The second sub-guide plate **1233** has a size suitable to be inserted into the hollow inside of the first sub-guide plate **1232**. The inside of the second sub-guide plate **1233** is hollowed along a direction along which the first sub-guide plate **1232** extends. The connection guide plate **1234** is accommodated in the hollow portion of the second sub-guide plate **1233**.

[0127] The second driving rail **1233***a* and the second driving block **1233***b* are installed in the hollow inside of the second sub-guide plate **1233**.

[0128] The second driving rail **1233***a* is installed on an inner surface of the second sub-guide plate **1233**. The second driving rail **1233***a* extends parallel with respect to the second sub-guide plate **1233**. The second driving block **1233***b* is slidably coupled to the second driving rail **1233***a*.
[0129] The second driving block **1233***b* is slidably coupled to the second driving rail **1233***a*. The second driving block **1233***b* can be pushed or pulled parallel with respect to the second sub-guide plate **1233** and moved on the second driving rail **1233***a* in conjunction with sliding of the display panel assembly **1500**.
[0130] The connection guide plate **1234** is fixed to the second driving block **1233***b*. Further, the connection guide plate **1234** can be accommodated in the second sub-guide plate **1233** or deployed to the outside of the second sub-guide plate **1233** in conjunction with forward and backward movements of the second driving block **1233***b*.
[0131] The second driving rail **1233***a* and the second driving block **1233***b* can be configured as linear motion guides.
[0132] Further, a second guide block **1233***c* is installed on an outer side surface of the second sub-guide plate **1233**.
[0133] The second guide block **1233***c* engages with the first driving rail **1232***a* at a position where the second guide block **1233***c* does not interfere with the first driving block **1232***b*. The second guide block **1233***c* includes a slit inwardly recessed in one surface facing the first driving rail **1232***a*. The second guide block **1233***c* is slidably inserted into the first driving rail **1232***a* through the slit. The second sub-guide plate **1233** can be stably deployed from the first sub-guide plate **1232** or accommodated into the first sub-guide plate **1232** by the second guide block **1233***c*.
[0134] The connection guide plate **1234** supports an end of the display panel assembly **1500**. The connection guide plate **1234** has a size suitable to be inserted into the hollow inside of the second sub-guide plate **1233** and is slidably installed in the hollow inside of the second sub-guide plate **1233**. The connection guide plate **1234** is fixed to the second driving block **1233***b*. Further, the connection guide plate **1234** can be accommodated in the second sub-guide plate **1233** or deployed to the outside of the second sub-guide plate **1233** in conjunction with forward and backward movements of the second driving block **1233***b*.
[0135] Further, a fixing piece **1234***a* for fixing the display panel assembly **1500** is provided on the connection guide plate **1234**. The fixing piece **1234***a* can be fixed to the display panel assembly **1500** by a bolt and a nut.
[0136] Further, a third guide block **1234***b* is installed on an outer side surface of the connection guide plate **1234**.
[0137] The third guide block **1234***b* engages with the second driving rail **1233***a* at a position where the third guide block **1234***b* does not interfere with the second driving block **1233***b*. The third guide block **1234***b* includes a slit inwardly recessed in one surface facing the second driving rail **1233***a*. The third guide block **1234***b* is slidably inserted into the second driving rail **1233***a* through the slit. The connection guide plate **1234** can be stably deployed from the second sub-guide plate **1233** or accommodated into the second sub-guide plate **1233** by the third guide block **1234***b*.
[0138] The opening and closing plate **1240** is coupled rotatably to one side of the guide body **1210** and connected to the guide driver **1220**. Further, the opening and closing plate **1240** is rotated (tilted) in conjunction with rotation of the guide support **1230** by the guide driver **1220**.
[0139] The opening and closing plate **1240** closes the opening **11** of the roof panel **10** while the guide support **1230** and the display panel assembly **1500** are accommodated on the roof panel **10**. Further, the opening and closing plate **1240** opens the opening **11** of the roof panel **10** to deploy the guide support **1230** and the display panel assembly **1500** to below the roof panel **10**.
[0140] The opening and closing plate **1240** extends along the intersection direction and has a size equal to or smaller than the opening **11** of the roof panel **10**.
[0141] A groove in which the insertion protrusions **1212** of the guide body **1210** are inserted is formed on one side of the opening and closing plate **1240**. The insertion protrusions **1212** are

inserted into the opening and closing plate **1240**, and the opening and closing plate **1240** is rotatably supported by the guide support **1230**. When the guide driver **1220** does not operate, the opening and closing plate **1240** is disposed parallel with respect to the roof panel **10**, and the opening **11** of the roof panel **10** is closed. Further, the roof panel **10** has an overall flat surface. Thus, the aesthetics of the interior of the vehicle can be improved.

[0142] The second link **1222***c* of the guide driver **1220** is fixed to an upper surface of the opening and closing plate **1240**. While the guide driver **1220** tilts the guide support **1230**, the opening and closing plate **1240** connected to the guide driver **1220** through the second link **1222***c* is also tilted. When the opening and closing plate **1240** is tilted, the opening **11** of the roof panel **10** is opened and a path for deploying the guide support **1230** and the display panel assembly **1500** to extend below the roof panel **10** is formed.

[0143] The sensor **1250** is installed on the guide body **1210** and senses the first link **1222***b*. The first sensor **1251** and the second sensor **1252** of the sensor **1250**, which are disposed at different positions from each other, sense the rotation link **1222***ba* which moves along a circumferential direction of a circle centering on the one side of the rotation link **1222***ba*. When the sensor **1250** is disposed to face the other side of the rotation link **1222***ba* being rotated, the sensor **1250** transmits a signal to the power generator. The rotation of the rotation link **1222***ba* is stopped by the power generator.

[0144] The sensor **1250** includes the first sensor **1251** and the second sensor **1252**.

[0145] The first sensor **1251** is disposed on a rotation path of the rotation link **1222***ba*. The first sensor **1251** is disposed on a movement path of the other side of the rotation link **1222***ba* revolving along the circumferential direction. When the first sensor **1251** senses the rotation link **1222***ba*, the first sensor **1251** transmits a sensing signal to the power generator. When the power generator receives the sensing signal from the first sensor **1251**, the power generator stops the supply of power to the motor **1221**.

[0146] The second sensor **1252** is disposed on a rotation path of the rotation link **1222***ba*. The second sensor **1252** is disposed higher than the first sensor **1251** on the movement path of the other side of the rotation link **1222***ba* revolving along the circumferential direction. When the second sensor **1252** senses the rotation link **1222***ba*, the second sensor **1252** transmits a sensing signal to the power generator. When the power generator receives the sensing signal from the second sensor **1252**, the power generator stops the supply of power to the motor **1221**.

[0147] The driving of the motor **1221** is stopped in response to the signals received by the first sensor **1251** and the second sensor **1252**. The first sensor **1251** and the second sensor **1252** can determine whether to drive the motor **1221** and restrict the degree of tilting (e.g., a tilting angle) of the guide support **1230**, the opening and closing plate **1240**, and the display panel assembly **1500**.

[0148] Meanwhile, in a range in which the rotation link **1222***ba* is not sensed by the first sensor **1251** and the second sensor **1252**, the motor **1221** can be driven freely. Further, the degree of tilting (e.g., the tilting angle) of the guide support **1230**, the opening and closing plate **1240**, and the display panel assembly **1500** can vary.

[0149] The roller unit **1300** supports sliding of the display panel assembly **1500**.

[0150] The roller unit **1300** includes at least one upper roller **1310** and at least one lower roller **1320**.

[0151] A plurality of upper rollers **1310** is provided and disposed in an upper area at a relative high position in the first support frame **1110**. The plurality of upper rollers **1310** is spaced apart from each other along the one direction. The plurality of upper rollers **1310** on the display panel assembly **1500** rotatably supports the display panel assembly **1500**.

[0152] A plurality of lower rollers **1320** is provided and disposed in a lower area at a lower position than the plurality of upper rollers **1310** in the first support frame **1110**. The plurality of lower rollers **1320** under the display panel assembly **1500** rotatably supports the display panel assembly **1500**.

[0153] The lower rollers **1320** include at least one first lower roller **1321** and at least one second lower roller **1322**.

[0154] The first lower roller **1321** is disposed to face the plurality of upper rollers **1310**. The first lower roller **1321** rotatably supports a flat portion of the display panel assembly **1500** which is not bent.

[0155] The second lower roller **1322** is disposed in a downwardly tilting direction with respect to a plurality of first lower rollers **1321**. The second lower roller **1322** can be tilted at an angle of 90° to 130°, such as 110°, with respect to a plurality of first lower rollers **1321**. The second lower roller **1322** together with the guide support roller **1231***e* rotatably supports the display assembly **1500**. The second lower roller **1322** under the display panel assembly **1500**, which is bent, rotatably supports the display panel assembly **1500**. Further, the guide support roller **1231***e* on the display panel assembly **1500**, which is bent, rotatably supports the display panel assembly **1500**.

[0156] Each of the plurality of upper rollers **1310** and the plurality of lower rollers **1320** is configured as an idle roller. Each idle roller may have a cylindrical or spherical shape.

[0157] FIG. **8**A is a perspective view illustrating a structure of a panel transfer unit according to an embodiment of the present disclosure. FIG. **8**B is an exploded perspective view illustrating the structure of the panel transfer unit according to an embodiment of the present disclosure. FIG. **9** is an exploded perspective view of a display panel assembly according to an embodiment of the present disclosure.

[0158] Referring to FIG. **1**A through FIG. **9**, the panel transfer unit **1400** is mounted on the mounting frame **1100** and serves to slide the display panel assembly **1500**.

[0159] The panel transfer unit **1400** includes a coupling plate **1410**, at least one support block **1420**, a rotary shaft **1430**, a driving motor **1440**, a carrier **1450**, a pair of supports **1460**, at least one linear motion (LM) guide **1470**, and a transfer bridge **1480**.

[0160] The coupling plate **1410** is mounted on the second support frame **1120** and fixed to the plurality of first support frames **1110**. The support block **1420**, a driving motor support member **1441**, and the pair of supports **1460** can be fixed onto the coupling plate **1410**.

[0161] The support block **1420** can be more than one. Each support block **1420** can be fixed onto the coupling plate **1410**. The support block **1420** can be spaced apart from each other along a sliding direction of the display panel assembly **1500**.

[0162] A bearing is provided inside the support block **1420**. A first end and a second end of the rotary shaft **1430** are rotatably installed on the bearing provided inside the support block **1420**. A pair of bearings serve to fix the rotary shaft **1430** at a predetermined position so as not to be eccentric. That is the pair of bearings may be colinear. The pair of bearings support a weight of the rotary shaft **1430** as it rotates and a load applied to the rotary shaft **1430**.

[0163] The rotary shaft **1430** can move the carrier **1450** in the sliding direction of the display panel assembly **1500** by means of the rotational driving force transmitted by the driving motor **1440**.

[0164] The rotary shaft **1430** can be configured as a ball screw. The rotary shaft **1430** serves to convert a rotational motion of the driving motor **1440** into a linear reciprocating motion of the carrier **1450** disposed on the rotary shaft **1430**. The rotary shaft **1430** is rotated by the driving motor **1440** and serves to linearly move the carrier **1450** toward one side or the other side of the support block **1420**. That is, the rotary shaft **1430** can be rotated by the drive motor **1440** and rectilinearly move the carrier **1450** between one end of the support blocks **1420** and the other end of the support blocks **1420**. Alternatively, other devices that convert rotary motion to linear motion, such as, a rack and pinion, could be used.

[0165] The driving motor **1440** is connected to the support block **1420** and serves to transmit a driving force to the rotary shaft **1430**.

[0166] The driving motor **1440** is connected to the rotary shaft **1430** through a coupler. The coupler holds the rotary shaft **1430** during movement of the carrier **1450** even when the concentricity (co-axis) of the rotary shaft **1430** is deviated and rotates the rotary shaft **1430** in a predetermined axial direction.

[0167] The driving motor **1440** connected to the other side of the support block **1420** is supported by the driving motor support member **1441** disposed between the driving motor **1440** and the coupler. The driving motor support member **1441** is fixed onto the coupling plate **1410**.
[0168] The driving motor **1440** can be connected to a power generator, such as an external power supply or a built-in battery and supplied with power. The driving motor **1440** generates a rotational driving force and transmits the rotational driving force to the rotary shaft **1430** configured as a ball screw.
[0169] The driving motor **1440** can rotate the rotary shaft **1430** depending on a position of a first LM block **1472** received by the power generator from a position sensor S.
[0170] The carrier **1450** is disposed on the rotary shaft **1430** and can linearly move in the sliding direction of the display panel assembly **1500** by means of the rotational driving force transmitted by the driving motor **1440**.
[0171] The pair of supports **1460** are disposed on opposite sides of the support block **1420**. The pair of supports **1460** are fixed onto the coupling plate **1410** along a movement direction of the carrier **1450**.
[0172] A plurality of position sensors S is provided and spaced apart from and parallel to each other along the one direction on one side of the support block **1420**. The plurality of position sensors S senses a position of the first LM block **1472** which moves together with the carrier **1450**. The plurality of position sensors S transmits the sensed position signal of the first LM block **1472** to the power generator.
[0173] The plurality of position sensors S serves as limit sensors to restrict movement of the carrier **1450** by sensing the position of the first LM block **1472**. Since the movement of the carrier **1450** is restricted by the plurality of position sensors S, it is possible to suppress damage or breakage of the panel transfer unit **1400** or the generation of noise.
[0174] The power generator can control (turn on or off) driving of the driving motor **1440** in response to the received position signal of the first LM block **1472**. The power generator can control the driving motor **1440** to rotate the rotary shaft **1430** in response to the received position signal of the first LM block **1472**. For example, the drive motor **1440** can be turned off when the first LM block **1472** reaches a predetermined position next to one of the position sensors S.
[0175] The LM guide **1470** supports sliding of the display panel assembly **1500**.
[0176] The LM guide **1470** includes a pair of first guide rails **1471** and a pair of first LM blocks **1472**.
[0177] The pair of first guide rails **1471** are fixed onto the pair of supports **1460**, respectively. The first guide rail **1471** serves as a path for moving the first LM block **1472**.
[0178] The pair of first LM blocks **1472** are movably coupled to the pair of first guide rails **1471**, respectively. A second transfer member **1482** is coupled onto the pair of first LM blocks **1472**. The pair of first LM blocks **1472** are coupled to the second transfer member **1482** and connected to the carrier **1450** through a first transfer member **1481** integrally formed with the second transfer member **1482** (see FIG. **8**A and FIG. **8**B). Therefore, the pair of first LM blocks **1472** operate in conjunction with movement of the carrier **1450**. For example, when the carrier **1450** reciprocates forwards and backwards along the one direction, the pair of first LM blocks **1472** connected to the carrier **1450** through the first transfer member **1481** and the second transfer member **1482** also reciprocate forwards and backwards in the one direction along the pair of first guide rails **1471** together with the carrier **1450**.
[0179] The transfer bridge **1480** is coupled onto the carrier **1450** and the first LM block **1472**. The transfer bridge **1480** moves the display panel assembly **1500** forwards and backwards.
[0180] The transfer bridge **1480** includes the first transfer member **1481** and the second transfer member **1482**.
[0181] The first transfer member **1481** is coupled onto the carrier **1450**. An interface board **1550** of the display panel assembly **1500** is coupled onto the first transfer member **1481**. Therefore, the

interface board **1550** of the display panel assembly **1500** as well as the pair of first LM blocks **1472** can slide by reciprocation of the carrier **1450**.

[0182] The second transfer member **1482** is formed integrally with the first transfer member **1481** and coupled onto the pair of first LM blocks **1472**. The second transfer member **1482** has a greater length than the first transfer member **1481** along the intersection direction, and has a greater thickness than the first transfer member **1481** in the vertical direction. A predetermined step is formed between the second transfer member **1482** and the first transfer member **1481**. The interface board **1550** is mounted at a step portion on the first transfer member **1481** having a smaller thickness than the second transfer member **1482**.

[0183] A connection panel **1560** of the display panel assembly **1500** is mounted on the second transfer member **1482** and can move together with the carrier **1450** in conjunction with movement of the carrier **1450**. Therefore, when the carrier **1450** reciprocates forwards and backwards along the one direction, the pair of first LM blocks **1472** also reciprocate. Further, the second transfer member **1482** coupled onto the pair of first LM blocks **1472** and the connection panel **1560** reciprocate forwards and backwards along the one direction. When the connection panel **1560** reciprocates forwards and backwards along the one direction, a display panel **1510** of the display panel assembly **1500** also moves.

[0184] The interface board **1550** of the display panel assembly **1500** is coupled onto the first transfer member **1481**. By movement of the carrier **1450** at the same time as movement of the display panel **1510**, the interface board **1550** slides together with the display panel **1510**.

[0185] The display panel assembly **1500** is slidably mounted on the mounting frame **1100**.

[0186] Referring to FIG. **1**A through FIG. **9**, the display panel assembly **1500** includes the display panel **1510**, a mid-cover **1520**, a plurality of back bars **1530**, a support panel **1540**, the interface board **1550**, and the connection panel **1560**.

[0187] The display panel **1510** can slide to below the roof panel **10** (i.e., the inside of the vehicle) through the opening **11**. The connection panel **1560** is coupled to one side of the display panel **1510** and connected to the second transfer member **1482** by the connection panel **1560**. When the carrier **1450** moves, the display panel **1510** can slide by means of a driving force transmitted through the first transfer member **1481** and the second transfer member **1482**.

[0188] The display panel **1510** includes a display element to display images, a driving element to drive the display element, and wiring lines to transmit various signals to the display element and the driving element. The display panel **1510** is usually disposed parallel with respect to the coupling plate **1410**, but a portion of the display panel **1510** can be bent by driving of the guide driver **1220**. While the display panel **1510** is bent, the display panel **1510** can be deployed to be tilted toward below the roof panel **10** through the opening **11** opened by tilting of the opening and closing plate **1240**.

[0189] The display panel **1510** is bent by the guide driver and the guide support and can be flexible to be slid while being bent.

[0190] Further, the display panel **1510** is connected to the interface board **1550** by the connection panel **1560**. The display panel **1510** receives an electrical signal from the interface board **1550** through the connection panel **1560**.

[0191] The mid-cover **1520** is disposed on one surface of the display panel **1510** and supports the display panel **1510**. The mid-cover **1520** can have a greater size than the display panel **1510** to protect the display panel **1510** from external impacts.

[0192] The mid-cover **1520** contains a malleable material so that plastic deformation does not occur in the mid-cover **1520** even when the display panel assembly **1500** is repeatedly slid. More specifically, the display panel **1510** can be bent by the guide member **1200**, and the mid-cover **1520** can contain a material having a yield stress equal to or greater than a stress applied to the mid-cover **1520** while the display panel **1510** is bent and slid. For example, the mid-cover **1520** can contain a material such as metal, rubber, plastic, or fabric. However, the material of the mid-cover **1520** is not limited thereto and can be variously modified depending on the design as long as it can satisfy property requirements, such as the amount of thermal deformation, radius of curvature, or rigidity.

[0193] The mid-cover **1520** can have a small thickness so that deformation does not occur in the mid-cover **1520** even when the display panel assembly **1500** is repeatedly slid.

[0194] A first adhesive layer AD**1** is disposed between the mid-cover **1520** and the display panel **1510**.

[0195] The first adhesive layer AD**1** serves to bond the mid-cover **1520** to the display panel **1510**. The first adhesive layer AD**1** is made of an adhesive material and contains a thermally curable or naturally curable adhesive. The first adhesive layer AD**1** can contain an optical clear adhesive (OCA), a pressure sensitive adhesive (PSA), etc., but is not limited thereto.

[0196] The plurality of back bars **1530** is disposed on one surface of the mid-cover **1520**.

[0197] The plurality of back bars **1530** supports the display panel **1510** and the mid-cover **1520**. The plurality of back bars **1530** can have a greater thickness than the mid-cover **1520** and protects the display panel **1510** and the mid-cover **1520** from external impacts.

[0198] The plurality of back bars **1530** contains a rigid material. The plurality of back bars **1530** can contain a metal material, such as steel use stainless (SUS) and Invar, or a plastic material. However, the material of the plurality of back bars **1530** is not limited thereto and can be variously modified depending on the design as long as it can satisfy property requirements, such as the amount of thermal deformation, radius of curvature, or rigidity.

[0199] A second adhesive layer AD**2** is disposed between the plurality of back bars **1530** and the mid-cover **1520**.

[0200] The second adhesive layer AD**2** serves to bond the mid-cover **1520** to the plurality of back bars **1530**. The second adhesive layer AD**2** is made of an adhesive material and contains an elastic material. The second adhesive layer AD**2** can contain a foam pad and elastic resin but is not limited thereto.

[0201] The support panel **1540** is coupled to ends of the plurality of back bars **1530**, and the fixing piece **1234***a* of the connection guide plate **1234** is coupled (e.g., screw-coupled) to an ed of the support panel **1540**. The support panel **1540** is supported by the connection guide plate **1234** through the fixing piece **1234***a*. Further, the mid-cover **1520** and the display panel **1510** are supported by the plurality of back bars **1530** coupled to the support panel **1540**. Therefore, when the display panel **1510** is deployed by the panel transfer unit **1400**, the connection guide plate **1234** fixed to the support panel **1540** (as well as the plurality of sub-guide plates **1232** and **1233** and the main guide plate **1231**) can also be deployed by means of a sliding force of the display panel **1510**.

[0202] The interface board **1550** is connected to the connection panel **1560**. The interface board **1550** is electrically connected to the display panel **1510** through the connection panel **1560**. The interface board **1550** can transmit and receive data, signals, or power for displaying images to and from the display panel **1510** through the connection panel **1560**. When power is supplied, the interface board **1550** can be connected to an image controller to control images displayed on the display panel **1510** connected through the connection panel **1560**.

[0203] The interface board **1550** is mounted on the first transfer member **1481** coupled onto the carrier **1450**. The interface board **1550** moves forwards and backwards along the one direction in conjunction with movement of the carrier **1450**.

[0204] The connection panel **1560** serves to electrically connect the interface board **1550** to the display panel **1510**. The connection panel **1560** connects the interface board **1550** to the display panel **1510**, which enables transmission and receipt of data, signals, or power for displaying images on the display panel **1510**. The connection panel **1560** contains a flexible material.

[0205] FIG. **10**A is a perspective view of the display panel assembly which is accommodated according to an embodiment of the present disclosure. FIG. **10**B is a bottom perspective view of the display panel assembly which is accommodated according to an embodiment of the present disclosure. FIG. **11**A is a perspective view of the display panel assembly which is bent according to an embodiment of the present disclosure. FIG. **11**B is a bottom perspective view of the display panel assembly which is bent according to an embodiment of the present disclosure. FIG. **12**A is a perspective view illustrating that a display panel of the display panel assembly is moved and exposed to the outside of a roof panel according to an embodiment of the present disclosure. FIG. **12**B is a bottom perspective view illustrating that the display panel of the display panel assembly is moved and exposed to the outside of the roof panel according to an embodiment of the present disclosure.

[0206] Hereinafter, a process of deploying the display panel assembly **1500** to the inside of the vehicle through the opening **11** of the roof panel **10** will be described with reference to FIG. **1**A through FIG. **12**B.

[0207] Referring to FIG. **10**A and FIG. **10**B, the display panel assembly **1500** is accommodated between the roof panel **10** and the outer plate of the vehicle. The display panel assembly **1500** is horizontally disposed along the one direction. In this state, the guide driver **1220** is not driven. Further, the guide support **1230** is disposed along the one direction. Further, the opening **11** of the roof panel **10** is partially covered by the opening and closing plate **1240** disposed horizontally with respect to the roof panel **10**. Furthermore, the other side of the rotation link **1222***ba* of the first link **1222***b* is disposed to face the first sensor **1251** among the plurality of sensors **1250**.

[0208] Referring to FIG. **11**A and FIG. **11**B, the guide driver **1220** is driven, and a part (i.e., a front end) of the display panel assembly **1500** is bent.

[0209] The motor **1221** of the guide driver **1220** operates to generate a rotational driving force. The rotational driving force generated by the motor **1221** is transmitted to the guide belt **1223** through the motor pulley **1221***a*. The rotational driving force transmitted to the guide belt **1223** is transmitted to the first link **1222***b*, the second link **1222***c*, and the third link **1222***d* through the link pulley **1222***a*.

[0210] As shown in FIG. **4**B, while one side of the rotation link **1222***ba* of the first link **1222***b* is supported by the guide body **1210**, the rotation link **1222***ba* is rotated to a position where the other side of the rotation link **1222***ba* interacts with the second sensor **1252**.

[0211] The connection link **1222***bb* and the support link **1222***bc* are rotated at the same rotation angle as the rotation link **1222***ba* in conjunction with rotation of the rotation link **1222***ba*.

[0212] When the connection link **1222***bb* is rotated, the second link **1222***c* connected through the third link **1222***d* is applied with pressure and tilts the opening and closing plate **1240** downwardly. While one side of the opening and closing plate **1240** is supported by the support protrusions **1211** of the guide body **1210**, the other side of the opening and closing plate **1240** is applied with pressure by the second link **1222***c* and tilted downwardly. When the opening and closing plate **1240** is tilted downwardly, the opening **11** is opened to provide a space through which the guide support **1230** is inserted into the opening **11**.

[0213] Further, when the support link **1222***bc* is rotated together with the connection link **1222***bb*, the guide support **1230** is also tilted downwardly. The main guide plate **1231** connected to the support link **1222***bc* (as well as the plurality of sub-guide plates **1232** and **1233** and the connection guide plate **1234**) is tilted downwardly and inserted into the opening **11**. In the process of tilting the main guide plate **1231**, the display panel assembly **1500** of which an end is supported by the connection guide plate **1234** is bent downwardly.

[0214] Further, in the process of tilting the guide support **1230**, the movable rail **1231***b* relatively movably coupled to the movable block **1231***c* of the main guide plate **1231** is pushed in conjunction with tilting (rotation) of the connection link **1222***bb* and the support link **1222***bc*. This is to bend the display panel assembly **1500** downwardly with a continuous length. By comparison between FIG. **4**A and FIG. **4**B, the support link **1222***bc* is located at an end on the other side of the main guide plate **1231** before the guide support **1230** is tilted. Further, the support link **1222***bc* is located adjacent to one side of the main guide plate **1231** after the guide support **1230** is tilted.

[0215] Furthermore, in the process of tilting the guide support **1230**, an end of the bent display panel assembly **1500** is mounted on the second lower roller **1322**. An upper surface of the display panel assembly **1500** mounted on the second lower roller **1322** comes into contact with the guide support roller **1231***e*.

[0216] Moreover, in the process of tilting the guide support **1230**, the end of the bent display panel assembly **1500** is disposed in parallel to the tilted opening and closing plate **1240**.

[0217] Referring to FIG. **12**A and FIG. **12**B, when the panel transfer unit **1400** operates, the display panel assembly **1500** is guided by the tilted guide member **1200** and deployed to below the roof panel **10** (i.e., the inside of the vehicle).

[0218] When the driving motor **1440** of the panel transfer unit **1400** operates, a rotational driving force is generated. The rotational driving force generated by the driving motor **1440** is transmitted to the carrier **1450** through the rotary shaft **1430**. The carrier **1450** slides and moves forwards in the one direction along the support block **1420**.

[0219] When the carrier **1450** moves forwards, the carrier **1450** pushes the transfer bridge **1480** along the one direction. The first LM block **1472** connected through the first transfer member **1481** and the second transfer member **1482** is guided along the first guide rail **1471** and slid in conjunction with forward movement of the carrier **1450**.

[0220] As shown in FIG. **4**C, when the transfer bridge **1480** moves in the one direction in conjunction with movement of the carrier **1450**, the connection panel **1560** coupled to the second transfer member **1482** (as well as the mid-cover **1520**, the display panel **1510**, the back bar **1530**, and the support panel **1540**) slides forwards. In the process of moving the display panel assembly **1500** forwards, the bent display panel assembly **1500** is slid by the tilted guide support **1230** at a tilting angle of the guide support **1230**.

[0221] Further, in the process of moving the display panel assembly **1500** forwards, the guide support **1230** is deployed by the support panel **1540** fixed to the connection guide plate **1234**. When the support panel **1540** of the display panel assembly **1500** moves forwards in a tilting direction, the connection guide plate **1234** slides in the tilting direction. The connection guide plate **1234** slides in the tilting direction along the second driving rail **1233***a*, and the second driving block **1233***b* fixed to the connection guide plate **1234** is disposed at an end of the second driving rail **1233***a*.

[0222] While the second driving block **1233***b* is disposed at the end of the second driving rail **1233***a*, the display panel assembly **1500** can move further forwards. In this case, the second driving block **1233***b* applies pressure to an inner surface of the second sub-guide plate **1233** to deploy the second sub-guide plate **1233** to the outside of the first sub-guide plate **1232**. In this way, the first sub-guide plate **1232** is deployed from the inside to the outside of the main guide plate **1231**.

[0223] The connection guide plate **1234**, the first sub-guide plate **1232**, and the second sub-guide plate **1233** are deployed in a parallel direction with respect to each other. Therefore, the display panel **1510** of which an end is supported by the connection guide plate **1234** extends in the tilting direction and is disposed as shown in FIG. **4**C.

[0224] Accordingly, the display panel assembly **1500** can be tilted and guided by the guide member **1200** toward below the roof panel **10** and disposed inside the vehicle.

[0225] The embodiments of the present disclosure can also be described as follows:

[0226] According to an aspect of the present disclosure, a slidable display device comprises [0227] a mounting frame, a display panel assembly movably coupled to the mounting frame, and a guide member which is tiltably coupled to the mounting frame, supports the display panel assembly, and guides movement of the display panel assembly, wherein the guide member is configured to bend at least a part of the display panel assembly while being tilted.

[0228] The guide member can be tilted between a first position and a second position, the display panel assembly can be horizontally disposed while the guide member is disposed at the first position, and at least a part of the display panel assembly can be bent and can be extended below the mounting frame while the guide member is disposed at the second position.

[0229] The guide member can include a guide body installed on one side of the mounting frame, a guide support which is connected to the display panel assembly and rotatably tilted with respect to the mounting frame, and a guide driver which is installed on the mounting frame and the guide body and can drive rotatably the guide support.
[0230] The guide member can further include an opening and closing plate which is coupled rotatably to one side of the guide body, connected to the guide driver, and can be rotated in conjunction with rotation of the guide support.
[0231] The guide driver can include a motor which is installed on the other side of the mounting frame and generates a driving force, a link assembly which connects the guide body to the guide support, and a guide belt which transmits the driving force of the motor to the link assembly.
[0232] The link assembly can include a first link which is connected to the guide belt and of which one side is rotatably installed on the guide body and the other side supports the guide support, a second link coupled to the opening and closing plate, and a third link which connects the first link to the second link.
[0233] The first link can include a rotation link of which one side is rotatably installed on the guide body and which can be tilted between a first position at which the display panel assembly is horizontally disposed and a second position at which the display panel assembly is extended below the mounting frame, a connection link of which one side is coupled to one side of the rotation link and a central portion is rotatably coupled to the third link and which can be tilted in conjunction with the rotation link, and a support link of which one side is fixed to the other side of the connection link and the other side supports the guide support.
[0234] The guide member can further include a plurality of sensors which is installed on the guide body and senses the rotation link being tilted.
[0235] The plurality of sensors can include a first sensor disposed at the first position, and a second sensor disposed at the second position, wherein the first sensor and the second sensor are disposed along a circumferential direction of a circle centering on the one side of the rotation link as a center axis.
[0236] The rotation link interacts with any one of the first sensor and the second sensor and a tilting angle of the rotation link can be restricted by one of the first sensor and the second sensor.
[0237] The first link, the second link, and the third link can constitute a pivotable four-bar link structure.
[0238] An end of the display panel assembly can be coupled to an end of the guide support, and as the guide support is applied with pressure by the display panel assembly applied with pressure by an end of the display panel assembly and increased in length, the guide support guides sliding of the display panel assembly.
[0239] The guide support can have a telescopic structure which can be increased in length.
[0240] A maximum deployment length of the guide support can be three times to four times greater than a height of the mounting frame.
[0241] The guide support can include a main guide plate which includes an accommodation groove on one side, a movable rail on both side portions intersecting the accommodation groove, and a movable block which is coupled to a support link of the guide driver, and can move along the movable rail, a sub-guide plate which includes a driving rail and a driving block movably coupled to the driving rail, has a hollow inside, and is slidably installed in the accommodation groove, and a connection guide plate to which the display panel assembly is fixed and which is slidably installed inside the sub-guide plate, wherein the movable block can be moved along the movable rail by rotation of the rotation link and moves the main guide plate in a direction opposite to a movement direction.
[0242] The slidable display device can further comprise a plurality of upper rollers spaced apart from and parallel to each other on one side of the mounting frame, and a plurality of lower rollers which is disposed under the plurality of upper rollers and rotatably supports the display panel

assembly together with the plurality of upper rollers.
[0243] The plurality of upper rollers and the plurality of lower rollers can be installed above the opening and closing plate on the mounting frame.
[0244] The plurality of lower rollers can include a plurality of first lower rollers disposed parallel with respect to the plurality of upper rollers, and a plurality of second lower rollers disposed in a downwardly tilting direction with respect to the plurality of first lower rollers, wherein a bending portion of the display panel assembly can be mounted on the plurality of second lower rollers in conjunction with tilting of the guide member.
[0245] The slidable display device can further comprise a panel transfer unit which is disposed on the mounting frame and coupled to the display panel assembly, and slides the display panel assembly.

Claims

1. A slidable display device comprises: a display panel assembly; a mounting frame; and a guide member tiltably connected to the mounting frame, the guide member being configured to support the display panel assembly as the display panel assembly moves between a first position and a second position with respect to the mounting frame, the guide member being configured to bend the display panel assembly as the display panel assembly moves from the first position to the second position so that a first portion of the display panel assembly is tilted downward.
2. The slidable display device of claim 1, wherein the first portion of the display panel assembly extends past the mounting frame at the second position.
3. The slidable display device of claim 1, wherein the guide member includes: a guide support connected to the display panel assembly; a guide body located at one end of the mounting frame; and a guide driver located at the mounting frame and the guide body.
4. The slidable display device of claim 3, wherein the guide driver further includes a link assembly, the link assembly configured to rotate the guide support and bend the display panel assembly.
5. The slidable display device of claim 4, wherein the guide member further comprises an opening and closing cover connected to the link assembly, the opening and closing cover being configured to rotate with the guide support.
6. The slidable display device of claim 5, wherein the guide driver further includes: a drive motor located at another end of the mounting frame opposite the one end of the mounting frame; a motor pulley connected to the drive motor; and a guide belt connecting the motor pulley to the link assembly, the guide belt being configured to rotate the link assembly.
7. The slidable display device of claim 6, wherein the link assembly includes: a first link rotatably connected to the guide support at a first end of the first link, the first link having a link pulley connected to the guide belt at a second end of the first link; a second link connected to the opening and closing plate; and a third link rotatably connecting the second link to the first link.
8. The slidable display device of claim 7, wherein the guide member further includes a pair of sensors spaced apart from each other, and wherein the link assembly includes a rotation link connected to the second end of the first link, the rotation link being configured to rotate between the pair of sensors as the display panel assembly moves between the first position and the second position.
9. The slidable display device of claim 3, wherein an end of the display panel assembly is connected to the guide support.
10. The slidable display device of claim 9, wherein the guide support includes: a main guide plate connected to the link assembly; and a connection guide plate connected to the end of the display panel assembly, the connection guide plate being configured to be slidable relative to the main guide plate.
11. The slidable display device of claim 10, wherein the guide support includes at least one sub-

plate connecting the main guide plate to the connection guide plate, and wherein the at least one sub-plate and the connection guide plate are configured to be received in a recess at a lower portion of the main guide plate.

12. The slidable display device of claim 10, wherein the guide support includes at least one guide support roller disposed at a lower side of the guide support, the at least one guide support roller being in contact with the display panel assembly.

13. The slidable display device of claim 12, further comprising: at least one upper roller at an upper portion of the one end of the mounting frame; and at least one lower roller at a lower portion of the one end of the mounting frame, wherein the at least one upper roller and the at least one guide support roller are configured to contact an upper surface of the display panel assembly, and wherein the at least one lower roller is configured to contact a lower surface of the display panel assembly.

14. The slidable display device of claim 1, further comprising a panel transfer unit connected to the mounting frame and the display panel assembly, the panel transfer unit being configured to apply a force to the display panel assembly to move the display panel assembly between the first position and the second position.

15. The slidable display device of claim 14, wherein the panel transfer unit includes: a drive motor; a rotary shaft connected to the drive motor; and a carrier configured to reciprocate between a first end of the rotary shaft and a second end of the rotary shaft, the carrier being connected to the display assembly panel.

16. A roof panel assembly for a vehicle, the roof panel assembly comprising: a roof panel having an opening; and a slidable display device connected to the roof panel assembly, the slidable display device including: a display panel assembly; a mounting frame connected to the roof panel; and a guide member tiltably connected to the mounting frame, the guide member being configured to support the display panel assembly as the display panel assembly moves between a first position above the roof panel and a second position in which a portion of the display panel assembly is tilted downward to extend through the opening of the roof panel.

17. The roof panel assembly of claim 16, wherein the guide member includes: a guide support connected to the display panel assembly; a guide body located at one end of the mounting frame; and a guide driver located at the mounting frame and the guide support to the guide body.

18. The roof panel assembly of claim 17, wherein the guide driver further includes a link assembly, the link assembly configured to rotate the guide support and bend the display panel assembly.

19. The roof panel assembly of claim 18, wherein the guide member further comprises an opening and closing cover connected to the link assembly, the opening and closing cover being configured to rotate with the guide support to at least partially open and close the opening of the roof panel.

20. The roof panel assembly of claim 17, further comprising a panel transfer unit connected to the mounting frame and the display panel assembly, the panel transfer unit being configured to apply a force to the display panel assembly to move the display panel assembly between the first position and the second position.